

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering An Autonomous Institute (Permanently affiliated to Savitribai Phule Pune University)	SEMESTER- VI
ASSIGNMENT- 6		
Name: Aditya Anil Naik PRN No:124B2B014 Class: TY B-4 Sub: FSDL		

1. Objective

The primary objective of this assignment is to develop a simple functional web application for buying and selling used items such as cars, bikes, and other products using Node.js and Express. The goal is to implement backend functionality, handle user input, and store data using a NoSQL database like MongoDB.

2. Learning Outcomes

After completing this assignment, I was able to understand how to build a backend application using Node.js and Express. I learned how to perform CRUD operations using MongoDB and how to connect the frontend with backend APIs. I also gained knowledge about handling form data, dynamic content rendering, and working with client-server architecture.

3. Brief Theory

Node.js is a JavaScript runtime environment that allows execution of JavaScript on the server side.

Express.js is a web framework used to simplify server-side development and API creation.

MongoDB is a NoSQL database that stores data in JSON-like documents, making it flexible and suitable for dynamic applications like e-commerce portals. In this project, the frontend interacts with the backend through HTTP requests, and the backend processes these requests, interacts with the database, and returns responses to the user.

4. Program Code and Output

Program Code: [Click here](#)

Output Screenshot:

Used Items Marketplace

Simple Node.js + Express assignment project

Available Items

Loaded from API/Items



Jupiter

Category	Condition
Scooter	Good
Price	Seller
Rs 50000	Atharva



Dulet

Category	Condition
Bike	Very Good
Price	Seller
Rs 100000	Aditya



BMW

Category	Condition
Car	Good
Price	Seller
Rs 5000000	Somesh

Sell an Item

Title

Category

-- Select --

Price (INR)

Seller Name

Phone

Condition

-- Select --

Image URL (optional)

<https://example.com/photo.jpg>

Add Listing

Item listed successfully.

Buy an Item

Select Item

-- Select available item --

Buyer Name

Buyer Phone

Place Order

Recent Orders

Refresh

Buy an Item

Select Item

-- Select available item --

Buyer Name

Buyer Phone

Place Order

Order placed successfully. Purchased Jupiter for Rs 50000.

Recent Orders

Refresh

Used Items Marketplace

Simple Node.js + Express assignment project

Available Items

Loaded from /api/items



Category
Scooter

Condition
Good

Price
Rs. 50000

Seller
Ashwika



Category
Bike

Condition
Very Good

Price
Rs. 160000

Seller
Aditya



Category
Car

Condition
Good

Price
Rs. 1800000

Seller
Somesh

5. Conclusion

In this assignment, I developed a simple used items sales portal using Node.js, Express, and MongoDB. I learned how to create backend APIs, manage data using a NoSQL database, and connect it with a frontend interface. This assignment helped me understand real-world web application development and improved my full-stack development skills.

